Consider an urn of w → white balls
b → black balls

Draw n balls at random without replacement

X = no. of white balls drawn out of n many balls

$$X \sim HGeom(w, b, n)$$

$X = k$; $0 \le k \le w$

$$P(X=k) = \frac{{}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

$0 \le k \le w$, $n \le b+k$

$n-k \le b$

L-6

Distribution of a r.v X

↓

PMF (for discrete) $f_X(j) = P(X=j)$

Named distributions (1) Bernoulli Distr.

$X \sim Bern(p)$

X = indicator of success/failure

$$X = \begin{cases} 1, & \text{if success with prob. } p \\ 0, & \text{fail} \end{cases}$$

PMF

$$p_X(0) = P(X=0) = 1-p$$
$$p_X(1) = p$$

Binomial distribution : n independent Bernoulli Trials

$X \sim Bin(n, p)$

X = no. of successes in n trials

PMF : $p_X(k) = P(X = k)$

$$^nC_k \cdot p^k (1-p)^{n-k}$$

Hypergeometric distribution :

$$X \sim HGeom(w, b, n)$$

X = No. of white balls in a sample of size n without replacement from a basket of

w white balls &
b black balls

$$p_X(k) = P(X=k) = \frac{{}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

Ex. $Y \sim HGeom(n, w+b-n, w)$

then PMF of X = PMF of Y.

$X \sim HGeo(w, b, n)$

$$P(Y=k) = \frac{{}^{n}C_k \cdot {}^{w+b-n}C_{w-k}}{{}^{w+b}C_w}$$

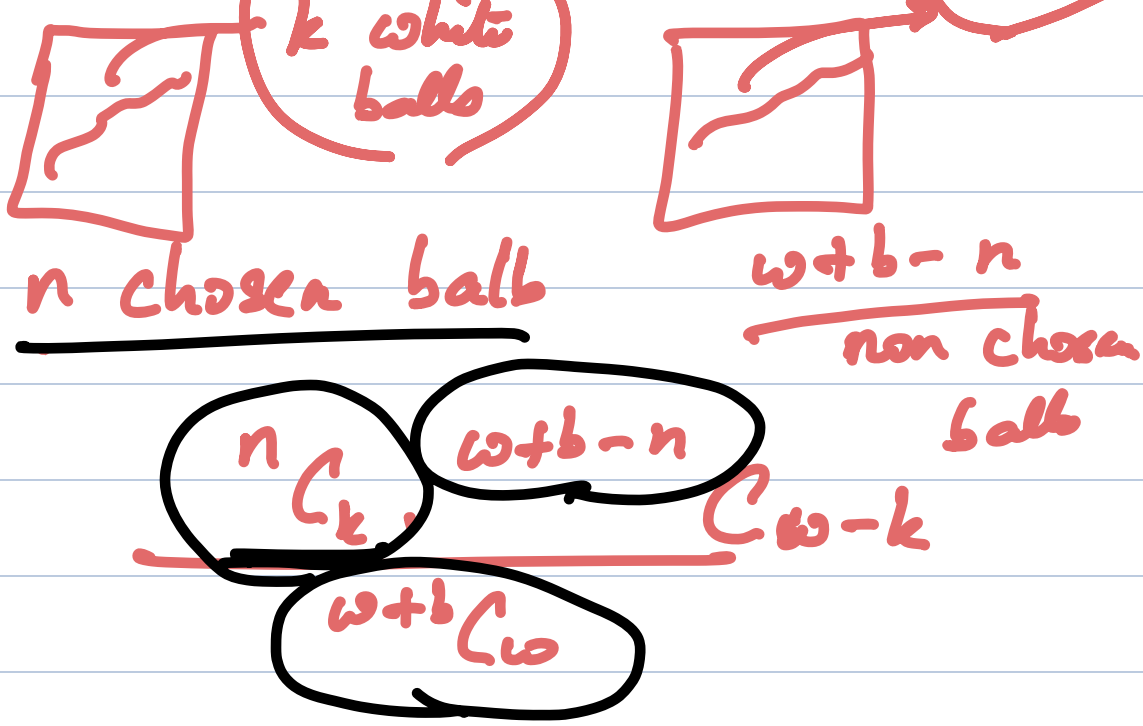
$$P(X=k) = \frac{{}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

$Z \sim \text{HGeom}(m+n,\ \underline{l-n},\ 2l-2m-z)$

PMF of

$Y \sim \text{HGeom}(\underbrace{n,\ w+b-n}_{},\ w)$

$w-k$



Discrete Uniform Distribution

$$S = \{x_1, x_2, \ldots, x_n\}$$

$$P(X = x_i) = \frac{1}{n} \quad \text{for all } 1 \le i \le n$$

$$P(X = x_i) = \begin{cases} \frac{1}{n}, & \text{if } x_i \in S \\ 0 & \text{o/w} \end{cases}$$

Geometric distribution

$$X \sim \text{Geom}(p)$$

X = # of independent Bernoulli trials until you get a success

$$P(X=j) = (1-p)^{j-1} \cdot p = p(1-p)^{j-1}$$

$1 \leq j < \infty$

CDF (Cumulative Distribution func^n)

$\boxed{X = x}$ → event

$\boxed{X \leq x}$ → event

$$F(x) = P(X \leq x) \quad ; x \in \mathbb{R}$$

x is a real no.

$\{HT, TH, TT, HH\}$

$X = \#$ of Tails

$$P(X \leq 2.5) = P(X=0) + P(X=1) + P(X=2) = 1$$

$F_X(2.5)$

Valid CDF

Any CDF F has the following properties:

(i) If $x_1 \leq x_2$, Then

$$F(x_1) \leq F(x_2)$$

$$A \subseteq B$$

$$P(A) \leq P(B)$$

$$\{X \leq x_1\} \subseteq \{X \leq x_2\}$$

(ii) F is right continuous

$$F(a) = \lim_{h \to 0} F(a+h)$$

$$F(a) = \lim_{x \to a^+} F(x)$$

(iii) x

$$\lim_{x \to \infty} F(x) = 1$$

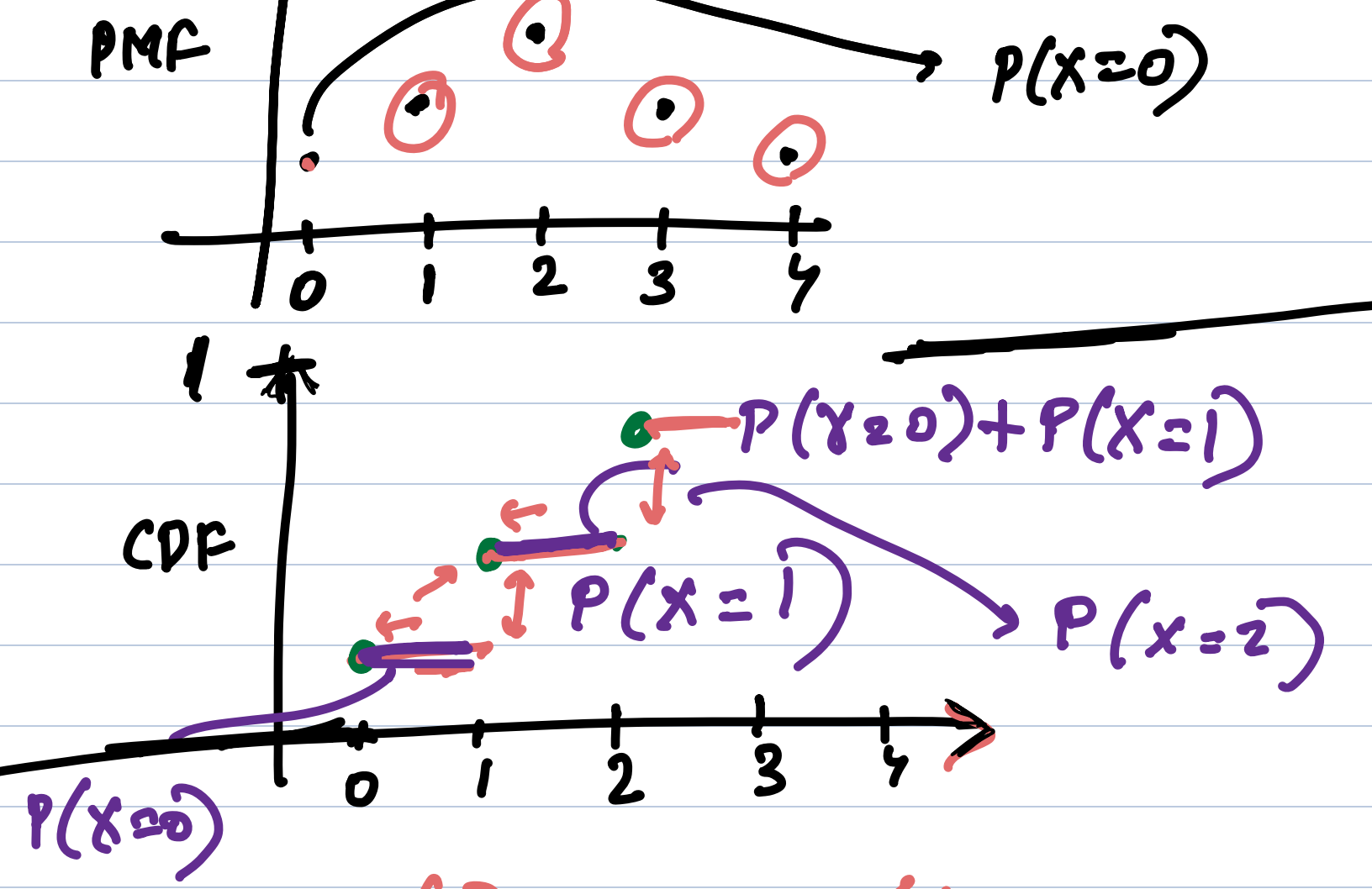
$$\lim_{x \to -\infty} F(x) = 0$$

$$F(x) = P(X \leq x)$$

x getting smaller & smaller

$$X > x$$

$$X \sim Bin(4, 1/2)$$



$F(x) \quad 0 \leq x < 1$

$= P(X \leq x)$

$F(1) = P(X \leq 1)$

PMF & CDF

$P(X \leq 3) = P(X=0) + X(X=1) + P(X=2) + P(X=3)$

$$P(1 < X \leq 3)$$

- $$\{X \leq 3\} = \{X \leq 1\} \cup \{1 < X \leq 3\}$$

$$P(X \leq 3) = P(X \leq 1) + P(1 < X \leq 3)$$

$$P(1 < X \leq 3) = P(X \leq 3) - P(X \leq 1)$$
$$= F(3) - F(1)$$

$$\boxed{P(a < X \leq b) = F(b) - F(a)}$$

$$P(X = b)$$

$$= F(b) - F(b-1)$$